Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LEARNER GUIDE

For

Tableau Data Prep Masterclass

Course Code: C181

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	4 October 2026	Initial release of Tableau Data Prep Masterclass (C181): two-day 9:30am–5:30pm schedule.	Dr. Alfred Ang

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

TABLE OF CONTENTS

How to Use This Guide..... 5
Course Information.....5
Learning Outcomes.....5
Before You Start.....5
Standard Validation Routine.....5
Topic 01 — Connecting to and Preparing Data in Tableau Desktop (23%).....7
Domain Concepts.....7
Create live connections and extracts.....7
Create and manage the data model.....8
Manage data properties.....9
Cleanse data with Tableau Prep Builder.....10
Lab 01 — Audit a Retail Sales Connection and Extract.....13
Lab 02 — Model Orders, Returns, Products and Targets.....15
Lab 03 — Union Monthly Files and Join Store Attributes.....17
Lab 11 — Cleanse a Messy Sales Extract with Tableau Prep Builder.....19
Topic 02 — Exploring and Analyzing Data in Tableau Desktop (37%).....22
Domain Concepts.....7
Create basic charts.....22
Organise data and apply filters.....24
Apply analytics to a worksheet.....25
Lab 04 — Build the Core Tableau Chart Portfolio.....27
Lab 05 — Create Groups, Sets, Hierarchies and Filters.....29
Lab 06 — Apply Calculations, Table Calculations and Parameters.....31
Lab 07 — Analyse Southeast Asia Logistics Geographically.....33
Topic 03 — Sharing Insights with Dashboards and Workbooks (25%).....35
Domain Concepts.....7
Format views for presentation.....35
Create and modify a dashboard.....36
View and share workbook data.....37
Lab 08 — Build an Interactive Service Operations Dashboard.....39
Lab 09 — Optimise, Package and Share an Executive Workbook.....41
Topic 04 — Understanding Tableau Concepts and Certification Preparation (15%).....43
Domain Concepts.....7
Understand dimensions and measures.....43
Understand discrete and continuous fields.....44
Understand aggregations.....44
Lab 10 — Capstone: Northstar Retail Executive Workbook.....46

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Quick Tableau Reference **48**
Certification Preparation **48**
Support **49**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

How to Use This Guide

The slide deck teaches concepts, comparisons and exam reasoning. This Learner Guide contains the detailed click-by-click procedures. Keep it open beside Tableau Desktop and work through the eleven lab folders in order. Each lab is self-contained: README.md, a realistic Excel workbook, starter.twb and an acceptance checklist.

Synthetic data: All names and records in the lab workbooks are fictional and generated for training. They are safe for classroom use and must not be represented as real organisational results.

Course Information

Course	Tableau Data Prep Masterclass
Course Code	C181
Duration	2 days · 15 instructional hours
Lab Root	labs/ — one folder per lab

Learning Outcomes

- LO1: Identify and connect appropriate data sources, then prepare trustworthy Tableau data models.
- LO2: Organise fields and create effective visual elements that answer business questions.
- LO3: Extract, combine and analyse data using relationships, joins, unions, filters and calculations.
- LO4: Build, format and share interactive dashboards and workbooks that communicate meaningful insights.

Before You Start

- Install or open Tableau Desktop. Tableau Public may be used for skills that do not require restricted connectors or private publishing.
- Download or clone the repository and keep the complete lab folder structure.
- Open each Excel workbook and read the Read Me, Data Dictionary and Control sheets before connecting Tableau.
- Create an evidence subfolder within the current lab folder for screenshots and exports.
- Use the supplied starter.twb as an empty shell, or create a new workbook if your Tableau version rejects an older workbook format.
- Never upload real personal or confidential workplace data to Tableau Public.

Standard Validation Routine

1. State the expected grain of every sheet or table.

2. Compare Tableau's row count and date range with the Excel Control sheet.

3. Check null or unmatched keys before building the first view.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

4. Verify data type, geographic role, default aggregation and number/date format.

5. Reconcile at least one total in Tableau with the workbook control total.

6. Save evidence before proceeding.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Topic 01 — Connecting to and Preparing Data in Tableau Desktop (23%)

Connections · extracts · metadata · relationships · joins · unions · field properties · Tableau Prep cleansing

Domain Concepts

Concept	Meaning
Connection strategy	Live prioritises source freshness; extracts prioritise speed, portability and predictable performance.
Model at the right layer	Relationships preserve table grain; joins create one physical row set; unions append rows.
Metadata is quality	Names, aliases, types, roles, formats and defaults determine how fields behave.
Clean before you model	Tableau Prep Builder repairs types, splits, nulls, duplicates and inconsistent members before analysis.
Validate first	Check row counts, null keys, duplicated grain and unmatched records before trusting a view.

Create live connections and extracts

1.1.1 — Create a live connection

Meaning	A live connection sends queries to the source during interaction.
Use when	Use when source freshness and central governance matter.
Exam clue	Current or real-time scenarios usually point to live.
Pitfall	Live is not automatically faster; network and query design still govern response time.

1.1.2 — Live versus extract

Meaning	The choice balances freshness, performance, portability and supported features.
Use when	Use an extract for offline analysis, acceleration or a stable reporting snapshot.
Exam clue	Select the mode that serves the scenario's priority.
Pitfall	An extract becomes stale until refreshed; a live connection may still cache results.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

1.1.3 — Create and refresh an extract

Meaning	An extract materialises selected data in Tableau's columnar .hyper format.
Use when	Filter rows or columns when the complete source is unnecessary.
Exam clue	Full refresh rebuilds; incremental refresh adds rows using a reliable key.
Pitfall	Incremental refresh does not update changed historical rows by itself.

1.1.4 — Save metadata in a .TDS

Meaning	A Tableau Data Source file stores connection details and field metadata.
Use when	Use it to standardise names, aliases, roles, folders and calculations.
Exam clue	TDS contains metadata; TDSX packages metadata with local files or extracts.
Pitfall	Do not confuse a data source definition with a packaged workbook (.twbx).

1.1.5 — Use multiple connections

Meaning	A data source may combine file, database or published connections.
Use when	Use when required tables do not reside in the same connection.
Exam clue	Connection is transport; data source is the analytical model exposed to sheets.
Pitfall	Cross-database joins can add query cost and connector limitations.

Create and manage the data model

1.2.1 — Add relationships

Meaning	Relationships describe how logical tables correspond without flattening first.
Use when	Use for multi-fact models or tables at different grains.
Exam clue	Tableau chooses context-aware joins and aggregation from fields in the view.
Pitfall	Incorrect cardinality assumptions can produce unexpected results.

1.2.2a — Add joins

Meaning	A join combines physical tables using matching clauses.
Use when	Use when row-level control and one flattened table are required.
Exam clue	Inner keeps matches; left preserves all left rows; right/full follow preservation

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

	logic.
Pitfall	Many-to-many joins can multiply rows and inflate measures.

1.2.2b — Add unions

Meaning	A union stacks records vertically from compatible tables.
Use when	Use for monthly or regional files with the same columns.
Exam clue	Wildcard unions can include similarly named tables or files.
Pitfall	Mismatched names create separate columns until they are merged.

1.2.3 — Choose join or relationship

Meaning	Relationships preserve independent grains; joins establish one physical grain.
Use when	Prefer relationships for flexible analysis; use joins for deliberate shaping.
Exam clue	Different grains or duplicated measures usually favour relationships.
Pitfall	A relationship line does not guarantee correct keys; validate it.

Manage data properties

1.3.1 — Rename a field

Meaning	A caption can be business-friendly without changing the source column.
Use when	Replace codes or database naming conventions with clear language.
Exam clue	Renaming affects Tableau metadata, not the source system.
Pitfall	Do not create ambiguous duplicate captions.

1.3.2 — Assign aliases

Meaning	Aliases replace displayed member labels while retaining original values.
Use when	Use for abbreviations or coded categories.
Exam clue	Aliases apply to discrete dimension members.
Pitfall	Aliases do not cleanse the source value.

1.3.3 — Assign geographic roles

Meaning	A geographic role tells Tableau how to geocode a field.
Use when	Use for country, state, city, postal code or

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

	custom location analysis.
Exam clue	Generated Latitude and Longitude appear after a valid role.
Pitfall	Ambiguous places need country or state context.

1.3.4 — Change a data type

Meaning	The type defines valid operations, parsing and visual encodings.
Use when	Correct imported IDs, dates, decimals and Boolean indicators.
Exam clue	A calendar-looking string is not a date until parsed correctly.
Pitfall	Conversion can create nulls where values are invalid.

1.3.5 — Set default properties

Meaning	Defaults control aggregation, number/date format, colour and sort.
Use when	Use when a field should behave consistently on new sheets.
Exam clue	Default aggregation can still be changed in a view.
Pitfall	Currency formatting does not convert currencies or units.

Cleanse data with Tableau Prep Builder

1.4.1 — Connect a Prep input and use the Data Interpreter

Meaning	Input steps bring in Excel, CSV, database and published sources; the Data Interpreter strips titles, notes and blank banding.
Use when	Use when spreadsheets carry cosmetic headers or merged title rows above the real table.
Exam clue	The input step is where sampling, field removal and input-level filters are set.
Pitfall	Interpreting an already-clean sheet can drop legitimate header rows; always review the result.

1.4.2 — Add a Clean step and read the profile pane

Meaning	The Clean step profiles each field, showing value distribution, nulls and outliers.
Use when	Use it first to see what is actually wrong before deciding on fixes.
Exam clue	Clicking a profile bar filters the data grid to those records.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Pitfall	Judging quality from the data grid alone hides the distribution problems the profile reveals.

1.4.3 — Fix data types and repair non-numeric text

Meaning	Currency, thousands separators and stray characters block numeric conversion.
Use when	Use a calculated field such as a REPLACE to strip commas before changing the type.
Exam clue	A field that refuses to become a number usually contains punctuation or spaces.
Pitfall	Converting directly to a number silently produces nulls where parsing fails.

1.4.4 — Apply cleaning operations

Meaning	Prep offers Remove Letters, Remove Numbers, Remove Punctuation, Trim Spaces, Remove Extra Spaces and case conversion.
Use when	Use to normalise names, codes and free-text categories.
Exam clue	Remove Extra Spaces collapses repeats; Remove All Spaces deletes every space.
Pitfall	Removing all spaces destroys multi-word names that still need to be readable.

1.4.5 — Split fields automatically and manually

Meaning	Automatic split infers a separator; custom split specifies the delimiter and number of columns.
Use when	Use to separate names, codes, dates or concatenated attributes.
Exam clue	Split creates new fields; the original is usually removed afterwards.
Pitfall	An inferred split breaks when the separator is inconsistent across rows.

1.4.6 — Filter and handle nulls and duplicates

Meaning	Filters can be by value, wildcard, range, calculation or null status; duplicates are found by profiling the row key.
Use when	Use to exclude test rows, negative sales, out-of-scope regions and empty keys.
Exam clue	Keep Only and Exclude are the fastest filters directly from the profile.
Pitfall	Filtering away nulls can silently drop valid records that merely lack one attribute.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

1.4.7 — Group and Replace to standardise members

Meaning	Group and Replace merges variants using manual selection, pronunciation, common characters or spelling similarity.
Use when	Use for misspelt city, product or customer names that should be one member.
Exam clue	Fuzzy matching proposes groups; the analyst confirms them.
Pitfall	Aggressive fuzzy matching can merge genuinely different members.

1.4.8 — Pivot columns to rows

Meaning	A Pivot step converts wide month or category columns into a tidy field-and-value pair.
Use when	Use when each period sits in its own column instead of its own row.
Exam clue	Pivoted output creates one row per original column value.
Pitfall	Leaving data wide forces one calculation per column and breaks date logic.

1.4.9 — Union, join and aggregate in a flow

Meaning	Union appends compatible files; Join matches on keys; Aggregate rolls rows to a coarser grain.
Use when	Use to combine monthly or country files and attach lookup attributes.
Exam clue	Mismatched field names appear in the union as separate columns until merged.
Pitfall	A join in Prep multiplies rows the same way a join in Desktop does.

1.4.10 — Create calculated fields and output the flow

Meaning	Prep calculations derive new columns; Output writes a CSV, Excel, .hyper extract or published data source.
Use when	Use to compute derived measures and publish a governed, refreshable clean table.
Exam clue	Running the flow re-executes every step in order against current input.
Pitfall	Outputting without re-running the flow ships stale results.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Lab 01 — Audit a Retail Sales Connection and Extract

Scenario	Northstar Retail needs a trustworthy weekly sales source before analysts build regional reporting.
Goal	Connect to Excel, validate field metadata, compare live and extract behaviour
You will produce	A validated Tableau data source, field-role audit, live/extract decision and refreshed .hyper extract
Tools	Tableau Desktop · Excel · live connection · extract · metadata
Folder	labs/lab-01-audit-a-retail-sales-connection-and-extract
Excel workbook	Lab-01-Northstar-Retail-Orders.xlsx
Suggested time	60 minutes

Detailed Step-by-Step Procedure

1. **Open Tableau Desktop and choose Microsoft Excel.**

   Select Lab-01-Northstar-Retail-Orders.xlsx from this lab folder.

2. **Inspect the Read Me and Data Dictionary worksheets first.**

   Confirm the scenario, expected grain and required fields before using Orders.

3. **Drag Orders to the data canvas and review the preview.**

   One row must represent one order line; compare the preview row count with the Control sheet.

4. **Correct field types.**

   Order Date and Ship Date are dates; Customer ID and Order ID are strings; Sales, Cost and Discount are decimals.

5. **Rename technical fields and add comments.**

   Use business-friendly captions while preserving source meaning.

6. **Assign geographic roles.**

   Set Country/Region, State/Province and City roles; resolve ambiguous locations with country context.

7. **Set default properties.**

   Format Sales, Cost and Profit as SGD; Discount as a percentage; default Profit aggregation as SUM.

8. **Create a worksheet named Connection Check.**

   Show record count, distinct orders, minimum date, maximum date and total sales.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

9. **Record the live/extract decision.**

   Use the decision table in the workbook and explain freshness, performance, portability and refresh implications.

10. **Create a Tableau extract.**

    Hide unused audit fields first, then extract all required rows.

11. **Run a full refresh and inspect extract properties.**

    Confirm the data-as-of timestamp and row count remain correct.

12. **Save the workbook as Lab-01-Connection-Audit.twbx.**

    Capture the Connection Check worksheet as evidence.

Acceptance Criteria

Pass standard: The source has the expected row count and date range, geographic fields are recognised, currency/percentage defaults are correct, and the extract refreshes without errors.

Complete acceptance-checklist.md in the lab folder. Submit the Tableau workbook, screenshots/exports requested in the README, one evidence-based insight and one data or design limitation.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Lab 02 — Model Orders, Returns, Products and Targets

Scenario	GreenBasket Grocers wants sales, returns and monthly targets analysed without duplicating measures across different table grains.
Goal	Create relationships and compare them with physical joins
You will produce	A four-table logical model with validated cardinality, unmatched-key checks and a join-risk comparison
Tools	Tableau Desktop · relationships · joins · referential integrity · grain validation
Folder	labs/lab-02-model-orders-returns-products-and-targets
Excel workbook	Lab-02-GreenBasket-Data-Model.xlsx
Suggested time	75 minutes

Detailed Step-by-Step Procedure

13. **Connect to Lab-02-GreenBasket-Data-Model.xlsx.**

    Read the Data Dictionary and Grain Audit sheets before modelling.

14. **Place Orders as the root logical table.**

    Confirm one row represents one order line.

15. **Relate Returns to Orders on Order ID and Line ID.**

    Use many-to-one or many-to-many only after checking the supplied uniqueness tests.

16. **Relate Products to Orders on Product ID.**

    Products should be one row per Product ID.

17. **Relate Targets using Month Start and Region.**

    Targets are monthly-region grain, not order-line grain.

18. **Review performance options.**

    State cardinality and referential integrity only when supported by evidence.

19. **Build a Model Validation worksheet.**

    Show Sales, Return Count and Target by Region without flattening the model.

20. **Check unmatched keys.**

    Use the Unmatched Keys sheet to identify intentional and erroneous exceptions.

21. **Create a duplicate copy of the data source for a join experiment.**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Join Orders to Returns in the physical layer and note the resulting row count.

22. **Compare relationship and join totals.**

    Explain why table grain and join type can change Sales and Return totals.

23. **Document the chosen model.**

    Add a caption that states each table grain and relationship key.

24. **Save as Lab-02-Relationship-Model.twbx.**

    Export the model canvas and validation worksheet screenshots.

Acceptance Criteria

Pass standard: Sales matches the control total, Return Count matches the Returns table, targets do not duplicate, and the learner can explain why relationships are safer for the mixed-grain model.

Complete acceptance-checklist.md in the lab folder. Submit the Tableau workbook, screenshots/exports requested in the README, one evidence-based insight and one data or design limitation.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Lab 03 — Union Monthly Files and Join Store Attributes

Scenario	MetroMart receives separate monthly transaction tabs plus a store master and needs one analysis-ready source.
Goal	Append compatible tables, merge mismatched fields and join a lookup
You will produce	A three-month union with provenance fields, merged columns, a validated store join and saved data-source metadata
Tools	Tableau Desktop · unions · wildcard patterns · joins · aliases · .tds/.tdsx
Folder	labs/lab-03-union-monthly-files-and-join-store-attributes
Excel workbook	Lab-03-MetroMart-Union-and-Join.xlsx
Suggested time	60 minutes

Detailed Step-by-Step Procedure

25. **Connect to Lab-03-MetroMart-Union-and-Join.xlsx.**

    Review the January, February and March sheet structures.

26. **Create a manual union of the three monthly sheets.**

    Confirm Tableau adds Table Name and Sheet provenance fields.

27. **Inspect null-heavy columns caused by a deliberate field-name mismatch.**

    February uses Sales Amount while other months use Net Sales.

28. **Merge mismatched fields.**

    Create one Net Sales field and verify February values are no longer null.

29. **Add the Stores sheet in the physical layer.**

    Left join on Store ID so all transactions remain.

30. **Validate the join.**

    Compare pre/post row counts and count unmatched Store IDs.

31. **Rename provenance fields.**

    Use Source Month and Source Sheet labels.

32. **Create aliases for store formats.**

    Display EXPRESS as Express and SUPERCENTER as Supercentre.

33. **Set default number and date formats.**

    Net Sales and Gross Profit are SGD; Transaction Date uses dd mmm yyyy.

34. **Build Monthly Union QA.**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Show monthly records, sales and unmatched-store counts.

35. **Save the reusable data source.**

    Create a .tds or .tdsx and state what is packaged.

36. **Save as Lab-03-Union-Join.twbx.**

    Capture the data-source canvas and QA view.

Acceptance Criteria

Pass standard: All three months appear, merged sales has no avoidable nulls, record counts match Control, the left join preserves transactions, and the saved source opens with metadata intact.

Complete acceptance-checklist.md in the lab folder. Submit the Tableau workbook, screenshots/exports requested in the README, one evidence-based insight and one data or design limitation.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Lab 11 — Cleanse a Messy Sales Extract with Tableau Prep Builder

Scenario	Harbourfront Trading exports raw transactions with currency text, inconsistent city spellings, split name fields, duplicate rows and wide monthly budget columns. Analysts need one governed, refreshable clean table before any Tableau Desktop analysis.
Goal	Build a repeatable Tableau Prep flow that profiles, cleans, reshapes and outputs an analysis-ready table
You will produce	A saved Tableau Prep flow (.tfl) with Input, Clean, Split, Group and Replace, Pivot, Union, Join and Output steps, plus a cleaned .hyper or CSV output and a documented change list
Tools	Tableau Prep Builder · profile pane · clean step · split · group and replace · pivot · union · join · output
Folder	labs/lab-11-cleanse-a-messy-sales-extract-with-tableau-prep-builder
Excel workbook	Lab-11-Harbourfront-Prep-Cleansing.xlsx
Suggested time	75 minutes

Detailed Step-by-Step Procedure

37. **Open Tableau Prep Builder and connect to Lab-11-Harbourfront-Prep-Cleansing.xlsx.**

    Start with the Q1 Transactions sheet; Q2 and Q3 are brought in and unioned at step 10. Read the Read Me, Data Dictionary and Control sheets first. If cosmetic title rows appear above the header, switch on Use Data Interpreter and confirm the corrected header row.

38. **Add a Clean step and study the profile pane before changing anything.**

    Record which fields show dirty types, nulls, outliers or too many distinct members. The profile, not the data grid, reveals distribution problems.

39. **Repair the Price Paid field so it can become a number.**

    Values contain currency symbols and thousands separators. Create a calculated field such as REPLACE(REPLACE([Price Paid],',',''),'$','') and only then change the type to Decimal Number; converting directly produces nulls.

40. **Apply cleaning operations to the free-text fields.**

    Use Trim Spaces and Remove Extra Spaces on Customer Name; note the difference from Remove All Spaces, which would destroy multi-word names. Use Remove Punctuation where codes carry stray characters.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

41. **Split the Customer Name field.**

    Use Automatic Split first, inspect the result, then use a custom split on the space delimiter if the inferred separator is inconsistent. Rename the outputs First Name and Last Name and remove the original field.

42. **Identify and remove duplicate rows.**

    Profile Transaction ID and confirm the distinct count against the Control sheet; exclude the duplicated records and re-check the count.

43. **Filter out invalid records.**

    Exclude rows with null Transaction ID and negative Sales using a calculation or range filter; state in a note how many rows each filter removed and why they are not legitimate data.

44. **Standardise city members using Group and Replace.**

    Apply pronunciation and spelling-similarity matching to merge misspelt city variants, then review every proposed group manually — fuzzy matching can merge genuinely different members.

45. **Pivot the wide monthly budget columns to rows.**

    Add a Pivot step on the Monthly Budget sheet, convert the twelve month columns into Month and Budget Amount fields, and set Month to a date type.

46. **Union the quarterly transaction sheets and merge mismatched fields.**

    Q2 uses Sales Amount while the others use Net Sales; merge them into one field and confirm no avoidable nulls remain. Keep the Table Name provenance field.

47. **Join the Region lookup and add calculated fields.**

    Inner join on Region Code, remove the duplicated key column, then add a Profit calculated field and verify it against the Control sheet.

48. **Reorder a step in the change list to see the effect, then restore it.**

    This proves the flow is an auditable, reorderable pipeline rather than a set of one-off clicks.

49. **Add an Output step and run the flow.**

    Write a .hyper extract or CSV named Lab-11-Clean-Output into this lab folder, then save the flow as Lab-11-Harbourfront-Cleansing.tfl.

50. **Connect Tableau Desktop to the cleaned output.**

    Build one validation worksheet showing record count, distinct customers, total sales and total profit, and reconcile against the Control sheet.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Acceptance Criteria

Pass standard: The flow runs end to end without errors; Price Paid is numeric, duplicates and invalid rows are removed, city members are standardised, budgets are pivoted long, the union has no avoidable nulls, the join does not multiply rows, and the output reconciles to the Control sheet.

Complete acceptance-checklist.md in the lab folder. Submit the Tableau workbook, screenshots/exports requested in the README, one evidence-based insight and one data or design limitation.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Topic 02 — Exploring and Analyzing Data in Tableau Desktop (37%)

Charts · groups · sets · hierarchies · filters · analytics · calculations

Domain Concepts

Concept	Meaning
Question before chart	Bars compare, lines show time, scatterplots show relationships and maps show location.
Structure narrows meaning	Groups, sets, hierarchies, sorts and filters determine the population compared.
Analytics adds context	Lines, totals, bins, calculations and parameters turn marks into evidence.
Order matters	Context, dimension, measure and table-calculation filters execute at different stages.

Create basic charts

2.1.1 — Create a bar chart

Meaning	Bars encode magnitude by length along a common baseline.
Use when	Use for ranking and comparing categories.
Exam clue	One dimension plus one measure is the canonical pattern.
Pitfall	Avoid 3-D effects, too many categories and unsorted ranking views.

2.1.2 — Create a line chart

Meaning	Lines connect ordered points to emphasise change over time.
Use when	Use for trends, seasonality and rate of change.
Exam clue	Continuous dates form axes; discrete date parts form headers.
Pitfall	Do not connect unrelated categories as a sequence.

2.1.3 — Create a scatterplot

Meaning	Two quantitative axes show the relationship between measures.
Use when	Use to assess correlation, clusters and outliers.
Exam clue	Add a dimension to Detail for one mark per entity.
Pitfall	Aggregation may collapse records into one mark.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

2.1.4 — Create a map

Meaning	Geographic fields generate spatial marks using Tableau geocoding.
Use when	Use when location contributes to the question.
Exam clue	Symbols compare points; fills compare areas; density shows concentration.
Pitfall	A bar chart is more accurate when geography adds no meaning.

2.1.5 — Create a combined-axis chart

Meaning	Multiple measures share one axis and Marks card.
Use when	Use when compatible units should be compared directly.
Exam clue	Measure Values and Measure Names are common building blocks.
Pitfall	Different units on one axis mislead.

2.1.6 — Create a dual-axis chart

Meaning	Two measures use independent axes with separate Marks cards.
Use when	Use to layer related measures with different scales or mark types.
Exam clue	Synchronise axes when measures share units.
Pitfall	Unsynchronised axes can imply artificial relationships.

2.1.7 — Create a stacked bar

Meaning	Segments divide each bar into contributions to a total.
Use when	Use for composition plus total comparison.
Exam clue	Colour defines the stack.
Pitfall	Middle segments lack a common baseline.

2.1.8 — Create a density map

Meaning	Overlapping points become a smooth concentration surface.
Use when	Use where hotspots matter more than individual points.
Exam clue	Intensity reflects local concentration and optional weighting.
Pitfall	Sparse data or smoothing can manufacture hotspots.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

2.1.9 — Crosstab and highlight table

Meaning	Text tables show exact values; colour adds pattern detection.
Use when	Use for lookup precision plus comparative scanning.
Exam clue	Measure Names and Measure Values can present several KPIs.
Pitfall	Too many cells overwhelm; filter and use meaningful scales.

Organise data and apply filters

2.2.1 — Create groups

Meaning	Groups combine members into higher-level categories.
Use when	Use to correct labels, merge small categories or define business groupings.
Exam clue	Create from marks, headers or the Data pane.
Pitfall	A manual group does not absorb new members automatically.

2.2.2 — Create sets

Meaning	A set divides members into IN and OUT using rules or selection.
Use when	Use for Top N, threshold cohorts and set actions.
Exam clue	Sets can be fixed, conditional, Top N or combined.
Pitfall	A set is not just a filter; it can remain visible as a category.

2.2.3 — Create hierarchies

Meaning	Hierarchies organise dimensions into a drill path.
Use when	Use Country > Region > City or Category > Product.
Exam clue	Plus/minus controls expand or collapse levels.
Pitfall	A hierarchy does not define data-model integrity.

2.2.4 — Add filters

Meaning	Filters restrict data or marks at specific order-of-operations stages.
Use when	Use to focus scope or support interaction.
Exam clue	Dimension, measure, context and table-calculation filters differ.
Pitfall	Filtering after aggregation is not filtering

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

	source rows.

2.2.5 — Add date filters

Meaning	Dates can be filtered by part, range, relative period or starting date.
Use when	Use for rolling windows and time scope controls.
Exam clue	Relative date filters move with the current date.
Pitfall	Discrete Month alone mixes the same month across years.

Apply analytics to a worksheet

2.3.1 — Manual and computed sorts

Meaning	Sorting controls the order of headers and marks.
Use when	Use manual order for business logic and computed order for ranking.
Exam clue	Nested sorts order members within panes.
Pitfall	Manual order can become stale as data changes.

2.3.2 — Add a reference line

Meaning	A line or band provides a benchmark for a scope.
Use when	Use for targets, averages, medians and ranges.
Exam clue	Scope can be table, pane or cell.
Pitfall	A table average can hide pane-level differences.

2.3.3 — Use quick table calculations

Meaning	Table calculations operate on marks already returned to the view.
Use when	Use running total, percent of total, difference, rank or moving average.
Exam clue	Compute Using defines addressing and partitioning.
Pitfall	Changing layout can change the calculation.

2.3.4 — Create bins and histograms

Meaning	Bins partition a measure into equal-width intervals.
Use when	Use a histogram for distribution shape and unusual ranges.
Exam clue	Bin size controls resolution.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Pitfall	Poor bin width can hide or invent patterns.

2.3.5 — Create calculated fields

Meaning	Calculated fields derive row-level, aggregate or table values.
Use when	Use for arithmetic, strings, dates and business rules.
Exam clue	Aggregate and non-aggregate expressions cannot be mixed directly.
Pitfall	Nulls propagate unless handled.

2.3.6 — Use parameters

Meaning	A parameter is a workbook value controlled by the user.
Use when	Use to swap measures, set thresholds or run what-if analysis.
Exam clue	A parameter acts only when referenced.
Pitfall	Parameters are not automatically row filters.

2.3.7 — Display totals

Meaning	Subtotals and grand totals summarise selected levels.
Use when	Use when audiences need components and overall results.
Exam clue	Total All Using can change aggregation.
Pitfall	Summed ratios and percentages often create invalid totals.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Lab 04 — Build the Core Tableau Chart Portfolio

Scenario	A consumer-electronics team needs views for ranking, trend, relationship, geography, composition and precise lookup.
Goal	Create and justify nine chart patterns from one realistic sales dataset
You will produce	Nine named worksheets: bar, line, scatterplot, symbol map, combined axis, dual axis, stacked bar, density map and highlight table
Tools	Tableau Desktop · Show Me · Rows/Columns · Marks · Measure Names/Values
Folder	labs/lab-04-build-the-core-tableau-chart-portfolio
Excel workbook	Lab-04-Orion-Electronics-Chart-Portfolio.xlsx
Suggested time	75 minutes

Detailed Step-by-Step Procedure

51. **Connect to Lab-04-Orion-Electronics-Chart-Portfolio.xlsx.**

    Use the Orders sheet and validate the Control totals.

52. **Build Sales by Category as a sorted bar chart.**

    Use length on a common baseline and meaningful labels.

53. **Build Monthly Sales and Profit as a line view.**

    Use a continuous Month date and annotate one material change.

54. **Build Discount vs Profit Ratio as a scatterplot.**

    Place Product on Detail and Category on Colour; identify clusters and outliers.

55. **Build Sales by City as a symbol map.**

    Use Sales on Size and Profit Ratio on Colour; fix any ambiguous geography.

56. **Build Sales and Cost on a combined axis.**

    Use Measure Names/Values only because both measures share SGD units.

57. **Build Sales vs Profit as a dual-axis view.**

    Use bars and a line, synchronise only if units and interpretation support it.

58. **Build Category Mix as stacked bars.**

    Limit colours and compare both total and composition.

59. **Build Order Density by Store Coordinates.**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Use generated Latitude/Longitude or supplied coordinates and explain smoothing.

60. **Build Region × Category as a highlight table.**

    Show exact Sales text with a sequential colour scale.

61. **Complete the Chart Choice worksheet.**

    For each view state the business question, visual encoding and one limitation.

62. **Save as Lab-04-Chart-Portfolio.twbx.**

    Export a contact sheet or screenshots of all nine views.

Acceptance Criteria

Pass standard: All nine chart types are present, each matches its stated question, axes and units are honest, maps resolve locations, and each view has a concise evidence-based insight.

Complete acceptance-checklist.md in the lab folder. Submit the Tableau workbook, screenshots/exports requested in the README, one evidence-based insight and one data or design limitation.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Lab 05 — Create Groups, Sets, Hierarchies and Filters

Scenario	HarbourLife Insurance needs a reusable segmentation view for advisers and portfolio managers.
Goal	Organise customer data and apply filters at the intended analytical stage
You will produce	Customer tier groups, Top-N and threshold sets, drill hierarchy, context filter and relative date controls
Tools	Tableau Desktop · groups · sets · hierarchy · context · date filters
Folder	labs/lab-05-create-groups-sets-hierarchies-and-filters
Excel workbook	Lab-05-HarbourLife-Customer-Segmentation.xlsx
Suggested time	60 minutes

Detailed Step-by-Step Procedure

63. **Connect to Lab-05-HarbourLife-Customer-Segmentation.xlsx.**

    Validate policy count, premium and claim totals.

64. **Create a Customer Tier group.**

    Combine micro and small-business codes into Emerging; keep Enterprise and Strategic separate.

65. **Create a Top 20 Customers set by SUM(Annual Premium).**

    Show IN/OUT instead of filtering it away.

66. **Create a High Claims set.**

    Use Claims Paid greater than the threshold documented on the Control sheet.

67. **Create a combined Risk Attention set.**

    Combine Top Customers and High Claims to identify commercially material risk.

68. **Build the Geography hierarchy.**

    Region > Country > City, then test expand/collapse.

69. **Build the Product hierarchy.**

    Product Family > Product Plan.

70. **Add an Active Policy filter.**

    Keep policies active as of the workbook's stated reporting date.

71. **Add a relative Renewal Date filter.**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Show renewals in the next 90 days.

72. **Make Region a context filter.**

    Observe how it changes the Top 20 set result and explain order of operations.

73. **Create an Adviser Worklist dashboard sheet.**

    Expose only purposeful filters and show set membership.

74. **Save as Lab-05-Segmentation.twbx.**

    Document one case where a group, set and filter would produce different outcomes.

Acceptance Criteria

Pass standard: The two hierarchies drill correctly, sets produce expected membership, relative dates use the reporting date, and the learner can explain how context changes Top-N evaluation.

Complete acceptance-checklist.md in the lab folder. Submit the Tableau workbook, screenshots/exports requested in the README, one evidence-based insight and one data or design limitation.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Lab 06 — Apply Calculations, Table Calculations and Parameters

Scenario	Crescent Hospitality wants profitability, occupancy and performance trends with user-controlled thresholds.
Goal	Create reliable row-level, aggregate and view-dependent analytics
You will produce	Calculated KPIs, running totals, percent-of-total, moving average, reference line and parameter-driven measure swap
Tools	Tableau Desktop · calculated fields · table calculations · parameters · reference lines · totals
Folder	labs/lab-06-apply-calculations-table-calculations-and-parameters
Excel workbook	Lab-06-Crescent-Hospitality-Analytics.xlsx
Suggested time	75 minutes

Detailed Step-by-Step Procedure

75. **Connect to Lab-06-Crescent-Hospitality-Analytics.xlsx.**

    Use the Daily Performance sheet and inspect KPI definitions.

76. **Create Gross Profit.**

    SUM([Room Revenue]) + SUM([F&B Revenue]) - SUM([Operating Cost]).

77. **Create Occupancy Rate.**

    SUM([Rooms Sold]) / SUM([Rooms Available]); format as percentage.

78. **Create RevPAR.**

    SUM([Room Revenue]) / SUM([Rooms Available]); verify its aggregation grain.

79. **Create Performance Band.**

    Use an IF/ELSEIF rule based on Occupancy Rate and Gross Profit.

80. **Build a monthly running revenue total.**

    Set Compute Using to Month and partition by Hotel.

81. **Build each hotel's percent of regional revenue.**

    Validate addressing and partitioning after rearranging the view.

82. **Add a 3-month moving average.**

    Explain the null edge at the start of each partition.

83. **Create a KPI Selector parameter.**

    Allow Room Revenue, Gross Profit, Occupancy and RevPAR.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

84. **Create a Selected KPI calculation.**

    Return the chosen measure with consistent aggregation.

85. **Add a parameter-driven target reference line.**

    Use a separate numeric Target parameter.

86. **Display totals and audit them.**

    Explain why averaging daily percentages differs from a ratio of summed numerators/denominators.

87. **Save as Lab-06-Advanced-Analytics.twbx.**

    Capture the parameter states and table-calculation settings.

Acceptance Criteria

Pass standard: Calculations match the Control sheet, table calculations keep intended addressing/partitioning, parameters affect the view, and totals use mathematically valid logic.

Complete acceptance-checklist.md in the lab folder. Submit the Tableau workbook, screenshots/exports requested in the README, one evidence-based insight and one data or design limitation.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Lab 07 — Analyse Southeast Asia Logistics Geographically

Scenario	Meridian Logistics wants to compare shipment volume, delay rates and port congestion across Southeast Asia.
Goal	Create symbol, filled and density maps with trustworthy geographic context
You will produce	Resolved geographic roles, three map types, tooltip context and a map-vs-bar decision note
Tools	Tableau Desktop · geographic roles · generated coordinates · density · tooltips
Folder	labs/lab-07-analyse-southeast-asia-logistics-geographically
Excel workbook	Lab-07-Meridian-SEA-Logistics.xlsx
Suggested time	60 minutes

Detailed Step-by-Step Procedure

88. **Connect to Lab-07-Meridian-SEA-Logistics.xlsx.**

    Review Shipments and Location Reference sheets.

89. **Assign geographic roles.**

    Country, State/Province and City must resolve within Southeast Asia.

90. **Use supplied Latitude/Longitude for ports where geocoding is ambiguous.**

    Treat them as geographic measures.

91. **Build Port Shipment Volume as a symbol map.**

    Use Shipment Count on Size and Delay Rate on Colour.

92. **Build Country On-Time Performance as a filled map.**

    Use a diverging scale centred on the target.

93. **Build Congestion Hotspots as a density map.**

    Weight by Delayed Shipments and state the smoothing implication.

94. **Create a Port Type shape map.**

    Use a small distinguishable shape set and keep a legend.

95. **Add informative tooltips.**

    Show Port, Country, Shipments, On-Time %, Avg Delay Hours and reporting period.

96. **Resolve unknown locations.**

    Use Edit Locations and document the correction.

97. **Create a ranked bar alternative.**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Compare accuracy and purpose with the filled map.

98. **Write one spatial insight and one limitation.**

    Base both on visible evidence.

99. **Save as Lab-07-Geographic-Analysis.twbx.**

    Export all maps and the bar comparison.

Acceptance Criteria

Pass standard: Locations resolve correctly, map types match the questions, tooltips carry operational context, colour scales are meaningful, and the learner can state when the bar chart is preferable.

Complete acceptance-checklist.md in the lab folder. Submit the Tableau workbook, screenshots/exports requested in the README, one evidence-based insight and one data or design limitation.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Topic 03 — Sharing Insights with Dashboards and Workbooks (25%)

Formatting · dashboards · actions · device layouts · stories · export · publish

Domain Concepts

Concept	Meaning
Format for meaning	Colour, type, shape, size and animation communicate data, not decoration.
Design interaction	Filters, highlighters, tooltips and actions answer predictable questions.
Layout is a system	Containers, hierarchy and device layouts govern alignment and reading order.
Share with intent	Package, publish or export while protecting sensitive detail.

Format views for presentation

3.1.1 — Use colour

Meaning	Colour can encode category, magnitude or emphasis.
Use when	Use hues for categories and ordered palettes for measures.
Exam clue	Legend type reveals discrete or continuous encoding.
Pitfall	Rainbow palettes and low contrast impair interpretation.

3.1.2 — Configure fonts

Meaning	Typography establishes hierarchy across titles, labels and tooltips.
Use when	Use consistent families and display-appropriate sizes.
Exam clue	Formatting applies at workbook, sheet, pane or element scope.
Pitfall	Tiny labels and excessive bolding reduce scanability.

3.1.3 — Format marks as shapes

Meaning	Shape is a categorical channel for point marks.
Use when	Use for a small number of distinguishable categories.
Exam clue	Custom shapes can be loaded and assigned.
Pitfall	Shape alone is weak for order and accessibility.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

3.1.4 — Configure viz animations

Meaning	Animations interpolate filter, sort and data changes.
Use when	Use to help audiences track moving marks.
Exam clue	Workbook formatting controls duration and style.
Pitfall	Motion should not be required for understanding.

3.1.5 — Change mark size

Meaning	Size can encode magnitude or improve legibility.
Use when	Use measure-driven size for proportional symbols.
Exam clue	Area, not radius, is perceived in circles.
Pitfall	Large marks overlap and small marks disappear.

3.1.6 — Show and hide legends

Meaning	Legends decode visual channels and can highlight marks.
Use when	Show when encoding is not self-evident; hide redundant cards.
Exam clue	Use the card menu or Analysis > Legends.
Pitfall	Removing a necessary legend makes marks ambiguous.

Create and modify a dashboard

3.2.1 — Add worksheets to a dashboard

Meaning	A dashboard composes sheets and objects into one canvas.
Use when	Create a clear hierarchy with the key view first.
Exam clue	Tiled items join containers; floating items use absolute position.
Pitfall	Adding every sheet creates noise.

3.2.2 — Add consumer interactions

Meaning	Filters, highlighters, tooltips and navigation support exploration.
Use when	Use controls that match likely questions and correct scope.
Exam clue	Apply to Worksheets defines controlled sheets.
Pitfall	Too many controls create contradictory states.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

3.2.3 — Add dashboard actions

Meaning	Actions respond to select, hover or menu events.
Use when	Use filter, highlight, parameter, set and URL actions.
Exam clue	Define source, target, event and clearing behaviour.
Pitfall	Wrong clearing behaviour can leave blank dashboards.

3.2.4 — Configure layouts and devices

Meaning	Containers and item hierarchy govern responsive alignment.
Use when	Use fixed/range/automatic sizes and device layouts.
Exam clue	Phone layouts can rearrange or omit objects.
Pitfall	Automatic sizing alone does not ensure mobile usability.

3.2.5 — Create a story

Meaning	A story sequences sheets or dashboards as annotated points.
Use when	Use for guided explanation rather than open exploration.
Exam clue	Each point stores a view state and caption.
Pitfall	Stories do not replace a coherent dashboard.

View and share workbook data

3.3.1 — Share a workbook

Meaning	Workbooks can be saved, packaged, exported or published.
Use when	Use TWBX for portability and Server/Cloud for governed collaboration.
Exam clue	A TWB may break if recipients cannot reach dependencies.
Pitfall	Publishing does not automatically make content public.

3.3.2 — View and export underlying data

Meaning	View Data exposes summary and permitted underlying rows.
Use when	Use to audit marks or export support records.
Exam clue	Selected marks and permissions define available data.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Pitfall	Exports can leak sensitive detail.

3.3.3 — Export to PowerPoint

Meaning	Tableau exports views as static PowerPoint images.
Use when	Use for offline briefings without interactivity.
Exam clue	Appearance is preserved, not filters or actions.
Pitfall	Label snapshots with a data-as-of date.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Lab 08 — Build an Interactive Service Operations Dashboard

Scenario	ApexCare's service director needs a daily dashboard for backlog, SLA risk, channel demand and team performance.
Goal	Compose worksheets, containers, filters, tooltips and dashboard actions
You will produce	A 1366×768 operational dashboard with KPIs, four views, scoped filters, highlight/filter actions and accessible tooltips
Tools	Tableau Desktop · dashboards · layout containers · filters · actions · tooltips
Folder	labs/lab-08-build-an-interactive-service-operations-dashboard
Excel workbook	Lab-08-ApexCare-Service-Operations.xlsx
Suggested time	90 minutes

Detailed Step-by-Step Procedure

100. **Connect to Lab-08-ApexCare-Service-Operations.xlsx.**

Review Tickets, Agents, SLA Targets and Control sheets.

101. **Create the four KPI calculations.**

Open Backlog, SLA Breach Rate, Median Resolution Hours and CSAT.

102. **Build Backlog by Team as a bar chart.**

Sort descending and use an SLA-risk highlight.

103. **Build Daily Ticket Trend as a line chart.**

Show Opened and Resolved tickets with consistent units.

104. **Build Channel × Priority as a highlight table.**

Use count on text and colour.

105. **Build Resolution vs CSAT as a scatterplot.**

Use Agent on Detail and Team on Colour.

106. **Create a fixed 1366×768 dashboard.**

Use tiled vertical and horizontal containers to establish hierarchy.

107. **Place KPI tiles and views.**

Keep the main decision view prominent and align padding consistently.

108. **Add Date, Team and Priority filters.**

Apply each only to relevant worksheets.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

109. **Add a Team filter action.**

Selecting a team bar filters the trend, matrix and scatterplot.

110. **Add a Priority highlight action.**

Hovering the matrix highlights related marks without removing context.

111. **Rewrite tooltips.**

Lead with the business meaning, then the key values and filter context.

112. **Test dashboard states.**

Use the acceptance checklist for default, selected, cleared and no-data states.

113. **Save as Lab-08-Service-Dashboard.twbx.**

Export dashboard PNG and action settings screenshots.

Acceptance Criteria

Pass standard: The dashboard loads in the target size, KPIs match Control, filters are scoped, actions clear correctly, no object overlaps, and a manager can answer the three stated operational questions.

Complete acceptance-checklist.md in the lab folder. Submit the Tableau workbook, screenshots/exports requested in the README, one evidence-based insight and one data or design limitation.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Lab 09 — Optimise, Package and Share an Executive Workbook

Scenario	Sentinel Energy must brief executives while giving analysts a governed interactive workbook.
Goal	Create device layouts, a story, exports and a governed sharing plan
You will produce	Desktop and phone layouts, three story points, TWB/TWBX comparison, PDF/PNG/PPT exports and a publishing checklist
Tools	Tableau Desktop · Device Designer · stories · TWBX · PDF/PNG/PPT · Server/Cloud
Folder	labs/lab-09-optimise-package-and-share-an-executive-workbook
Excel workbook	Lab-09-Sentinel-Energy-Executive-Sharing.xlsx
Suggested time	60 minutes

Detailed Step-by-Step Procedure

114. **Connect to Lab-09-Sentinel-Energy-Executive-Sharing.xlsx.**

    Review Energy Performance and Governance Notes.

115. **Build the Executive Performance dashboard.**

    Show demand, cost, renewable share and outage risk.

116. **Apply the presentation style guide.**

    Use accessible colours, readable type, consistent numbers and concise legends.

117. **Configure viz animations.**

    Use subtle animation for a year or scenario change and verify the final state remains understandable.

118. **Create a phone layout.**

    Reorder to KPI > key trend > exception list; remove nonessential views.

119. **Test multiple device previews.**

    Confirm legibility, touch targets and no horizontal scrolling.

120. **Create a story with three points.**

    Situation, evidence and recommended action; give every point a decision-oriented caption.

121. **Save a .twb and inspect dependencies.**

    Explain why a recipient without the source may see broken connections.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

122. **Save a .twbx.**

Confirm the Excel data is packaged and compare file size.

123. **Export PDF, PNG and PowerPoint.**

Record which interaction and freshness capabilities are lost.

124. **Draft a Server/Cloud publish plan.**

Specify project, permissions, refresh schedule, owner and certification status.

125. **Audit underlying-data exposure.**

Test View Data and export permissions against the Governance Notes.

126. **Save as Lab-09-Executive-Sharing.twbx.**

Submit both layouts, story, exports and the sharing decision table.

Acceptance Criteria

Pass standard: Desktop and phone layouts are usable, the story has three coherent points, TWBX is portable, static exports are labelled with data-as-of context, and the publish plan protects underlying data.

Complete acceptance-checklist.md in the lab folder. Submit the Tableau workbook, screenshots/exports requested in the README, one evidence-based insight and one data or design limitation.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Topic 04 — Understanding Tableau Concepts and Certification Preparation (15%)

Dimensions · measures · discrete · continuous · aggregation · exam strategy

Domain Concepts

Concept	Meaning
Role and type differ	Dimension/measure is analytical role; discrete/continuous is display behaviour.
Blue and green	Blue discrete pills make headers; green continuous pills make axes.
Aggregation follows grain	Measures aggregate at the level defined by dimensions.
Reason from consequences	Use pill colour, shelf placement, grain and order of operations.

Understand dimensions and measures

4.1.1 — What dimensions contain

Meaning	Dimensions usually contain names, categories, dates and locations.
Use when	Use them to segment, label, filter and define detail.
Exam clue	They often create headers and partitions.
Pitfall	A numeric code is not automatically a measure.

4.1.2 — What measures contain

Meaning	Measures usually contain quantitative values for aggregation.
Use when	Use for totals, averages, counts, ratios and axes.
Exam clue	Green measures create axes; blue measures can create headers.
Pitfall	Percentages and prices are not always additive.

4.1.3 — Dimensions versus measures

Meaning	Dimensions answer who/what/where/when; measures answer how much/how many.
Use when	Use both to define grain and numerical result.
Exam clue	Converting role changes partitioning and aggregation.
Pitfall	Do not confuse role with data type.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Understand discrete and continuous fields

4.2.1 — How discrete fields display

Meaning	Discrete fields create headers and partitions.
Use when	Use for named groups, panes and exact date parts.
Exam clue	Blue pills mean discrete behaviour.
Pitfall	Discrete does not mean text.

4.2.2 — How continuous fields display

Meaning	Continuous fields create axes over an unbroken range.
Use when	Use for quantitative scales and temporal trends.
Exam clue	Green pills mean continuous behaviour.
Pitfall	Continuous does not mean measure.

4.2.3 — Discrete date parts vs values

Meaning	Date parts classify recurring members; date values preserve chronology.
Use when	Use continuous dates for timelines and parts for seasonal grouping.
Exam clue	Blue MONTH shows Jan-Dec; green MONTH spans time.
Pitfall	Month-only can aggregate multiple years together.

Understand aggregations

4.3.1 — Default aggregation

Meaning	Measures aggregate to the current grain, commonly using SUM.
Use when	Change to AVG, MIN, MAX, COUNT or another appropriate operation.
Exam clue	The pill shows the active aggregation, such as SUM(Sales).
Pitfall	Number format and aggregation are separate.

4.3.2 — How dimensions change aggregation

Meaning	Each dimension partitions data into a finer detail.
Use when	Use dimensions deliberately to answer the required breakdown.
Exam clue	SUM(Sales) by Region differs from Region and Customer.
Pitfall	Unexpected totals may signal duplicated model rows or unintended dimensions.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Lab 10 — Capstone: Northstar Retail Executive Workbook

Scenario	Northstar Retail's COO needs an executive workbook that reconciles sales, returns, inventory and regional targets, then communicates actions across desktop and mobile.
Goal	Integrate all four exam domains and complete certification-readiness evidence
You will produce	A validated multi-table model, eight analytical worksheets, executive dashboard, phone layout, story, packaged workbook and exam-domain checklist
Tools	Tableau Desktop · relationships · calculations · charts · dashboards · stories · TWBX
Folder	labs/lab-10-capstone-northstar-retail-executive-workbook
Excel workbook	Lab-10-Northstar-Retail-Capstone.xlsx
Suggested time	100 minutes

Detailed Step-by-Step Procedure

127. **Connect to Lab-10-Northstar-Retail-Capstone.xlsx.**

    Read the Executive Brief, Data Dictionary, table grains and Control totals.

128. **Build the logical data model.**

    Relate Orders, Returns, Products, Stores and Monthly Targets with documented keys and grains.

129. **Validate the source.**

    Check row counts, unmatched keys, duplicated sales, date range and control totals.

130. **Create core calculations.**

    Net Sales, Gross Profit, Profit Ratio, Return Rate, Target Attainment and Days to Ship.

131. **Demonstrate field concepts.**

    Document dimensions/measures, discrete/continuous date behaviour and aggregation changes.

132. **Build eight worksheets.**

    KPI text table, regional bar, monthly line, product scatter, map, highlight table, return set analysis and target reference view.

133. **Use groups, sets and a hierarchy.**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Group store formats, create a Top-20 Products set and Region > Country > Store hierarchy.

134. **Apply analytics.**

Add a running total, percent of total, reference line, histogram and parameter-driven measure swap.

135. **Build Executive Overview dashboard.**

Use KPI tiles, four priority views, purposeful filters and at least two actions.

136. **Create a phone layout.**

Prioritise KPI, trend and exception list for the smaller screen.

137. **Create a three-point story.**

Performance, diagnostic evidence and recommended action.

138. **Package and export.**

Save TWBX; export PDF and PowerPoint snapshots labelled with the reporting period.

139. **Complete the Exam Domain Matrix.**

For every official objective, cite the workbook sheet or lab evidence that proves readiness.

140. **Run the acceptance checklist.**

Test default, filter, action, clear, mobile, export and underlying-data states.

141. **Save as Lab-10-Northstar-Capstone.twbx.**

Submit the workbook, exports, validation evidence and readiness matrix.

Acceptance Criteria

Pass standard: Control totals reconcile, the model avoids duplicated measures, all eight worksheets and required analytics exist, dashboard/actions work on desktop and phone, exports are current, and every official objective has evidence.

Complete acceptance-checklist.md in the lab folder. Submit the Tableau workbook, screenshots/exports requested in the README, one evidence-based insight and one data or design limitation.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Quick Tableau Reference

Term	Meaning
Live connection	Queries source on interaction; prioritises freshness.
Extract (.hyper)	Local analytical snapshot; refresh to update.
.twb / .twbx	Workbook XML / packaged workbook with local dependencies.
.tds / .tdsx	Data-source metadata / packaged data source.
Relationship	Logical, context-aware combination preserving table grains.
Join	Physical row combination with fixed join type and clauses.
Union	Vertical append of similarly structured rows.
Dimension / Measure	Analytical role: slice/label versus quantify/aggregate.
Discrete / Continuous	Display behaviour: blue headers versus green axes.
Table calculation	View-dependent calculation over returned marks.
Parameter	Single user-controlled workbook value; requires a consumer calculation/object.
Dashboard action	Filter, highlight, URL, parameter or set response to user events.

Certification Preparation

- Work through every official objective in Topics 1–4 and explain the result without relying on menu memory.
- Repeat all eleven labs in a clean workbook; use the acceptance tests to prove correctness.
- Practise recognising pill colour, field role, view grain and filter order from screenshots and scenarios.
- Read the official exam guide for current exam format, product version, timing and policies.
- Use Tableau's official free training videos for connection, workspace, concepts, dashboards, mapping and calculations.

Official exam guide: https://help.salesforce.com/s/articleView?id=005298988&type=1&language=en_US

Official Tableau training: https://www.tableau.com/learn/training

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Support

Course labs and materials: https://github.com/tertiarycourses/C181-Tableau-Data-Prep-Masterclass

Support: enquiry@tertiaryinfotech.com · +65 6100 0613 · www.tertiarycourses.com.sg

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg